



PALADIN
BLOCKCHAIN SECURITY

Smart Contract Security Assessment

Final Report

For LayerZero
(MessageLib Differential Audit)

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1 Overview

The MessageLib contracts at commit 6356f0a on the public LayerZero-v2 (<https://github.com/LayerZero-Labs/LayerZero-v2>) repository descend from the contracts audited by Paladin in December 2023.

Between the audit preliminary commit and the final resolution, the codebase underwent a significant architectural refactor - monolithic contracts were split along send/receive boundaries, the "Verifier" terminology was replaced with "DVN" and several standalone config files were absorbed into base contracts.

Result: All security-critical logic, including the fixes introduced during the audit, is the same at 6356f0a. The changes between the final audited commit (6ef8a4b) and the target commit are purely cosmetic: license headers (BUSL-1.1 -> LZBL-1.2), pragma versions (0.8.22 / ^0.8.22 -> ^0.8.20) and error name prefixes (e.g. InvalidConfigType() -> LZ_ULN_InvalidConfigType()). No functional code has been changed.

1.1 Summary

Original Audit	Paladin LayerZeroMessageLib Report (https://drive.google.com/file/d/13o1g86gF0kQf5C9fkWadgL4KWk5awYBv/view?pli=1), 15 December 2023
Contracts Audited	UltraLightNode302, UlnBase, UlnConfig, MessageLibBase, MessageLibBaseE2, OutboundConfig, UltraLightNode301, MessageLibBaseE1, TreasuryFeeHandler, AddressSizeConfig, UlnOptions, VerifierOptions
Target Commit	6356f0a (https://github.com/LayerZero-Labs/LayerZero-v2/commit/6356f0a31f2125267420e7bf8403b59a55459aa3)

1.2 Scope

- Original audit scope - twelve contracts across packages/layerzero-v2/evm/messagelib/contracts/ on the private monorepo. These underwent a major split/rename during the audit (see Section 4), resulting in 18+ descendant files at the target commit.
- Diff target - commit 6356f0a (<https://github.com/LayerZero-Labs/LayerZero-v2/commit/6356f0a31f2125267420e7bf8403b59a55459aa3>) on the public LayerZero-v2 repository.

1.3 Commit Timeline

Phase	Repository	Commit	Path
bddcada	v1.5.33	monorepo (private)	Preliminary
97a0bbb	v1.5.44	monorepo (private)	Resolution #1
faf36d0	v1.5.64	monorepo (private)	Resolution #2
6ef8a4b	v1.5.71	monorepo (private)	Resolution #3 - Final
6356f0a	-	LayerZero-v2 (public)	Target commit

1.4 Rename / Split Mapping

A large architectural refactor occurred between Resolution #1 and Resolution #2, splitting monolithic contracts along the send/receive boundary and absorbing standalone config files into base contracts.

Audited Name	Name(s) at 6356f0a	Change
UltraLightNode302	SendUln302 + ReceiveUln302	Split into send and receive
UlnBase	UlnBase + SendUlnBase + ReceiveUlnBase	Rewritten (Verifier->DVN), three-way split
UlnConfig	*(absorbed into UlnBase)*	Config logic merged into base contract
MessageLibBase	MessageLibBase	Gutted to simple endpoint/localEid container (163->21 lines)
MessageLibBaseE2	SendLibBaseE2 + ReceiveLibBaseE2	Split into send and receive
OutboundConfig	*(absorbed into SendLibBase)*	Merged into new SendLibBase
UltraLightNode301	SendUln301 + ReceiveUln301	Split into send and receive
MessageLibBaseE1	SendLibBaseE1 + ReceiveLibBaseE1	Split into send and receive
TreasuryFeeHandler	TreasuryFeeHandler	Same name, custom errors added
AddressSizeConfig	AddressSizeConfig	Same name (new file, first appears at faf36d0)
UlnOptions	UlnOptions	Same name, rewritten with custom errors
VerifierOptions	DVNOptions	Renamed

Fee logic originally in MessageLibBase was extracted to a new SendLibBase contract. The SafeCall library was introduced at Resolution #2 for treasury interactions.

1.5 Security Fixes During the Original Audit

The following security-relevant changes were made across the three resolution phases. All fixes remain intact at 6356f0a.

Preliminary (bddcada) -> Resolution #1 (97a0bbb):

- Refactored verifier job assignment to use AssignJobParam struct
- Improved UlnConfig validation: added isSupportedEid check, threshold > 0 requirement when optional verifiers are present
- Cached altFeeToken as immutable in MessageLibBaseE2
- Added supportsInterface to UltraLightNode302/301
- UlnOptions: added `_options.length >= 2` validation, `workerId > 0` check

Resolution #1 (97a0bbb) -> Resolution #2 (faf36d0):

- Complete send/receive split architecture
- Complete Verifier -> DVN terminology rename
- UlnConfig absorbed into UlnBase (config lives directly in the base contract)
- OutboundConfig absorbed into SendLibBase
- Custom errors replaced all `require()` + string patterns
- SafeCall library introduced for treasury interactions
- `NIL_DVN_COUNT` and `NIL_CONFIRMATIONS` sentinel values added

Resolution #2 (faf36d0) -> Resolution #3 / Final (6ef8a4b):

- Critical fix in ReceiveUlnBase: hashLookup changed from a bare uint64 to a Verification struct (bool submitted, uint64 confirmations). This eliminates zero-confirmation ambiguity - a submitted hash with zero required confirmations is now distinguishable from an unsubmitted hash.
- `treasuryNativeFeeCap` changed from immutable to mutable (down-only via owner) in SendLibBase
- `IUltraLightNode` -> `IReceiveUlnE2` interface rename in ReceiveUln302
- DVNOptions: explicit variable initializations

IMPORTANT: The Verification struct fix (Resolution #3) addresses the highest-severity finding of the audit. Its presence at 6356f0a was confirmed - ReceiveUlnBase stores mapping(bytes32 => mapping(bytes32 => Verification)) with the correct two-field struct.

1.6 Cross-References

Worker.sol is in scope for this audit but is covered more thoroughly in the DVN Diff Audit Report, where its functional additions (withdrawToken, supportedOptionTypes, hasAcl) are analyzed.

DVN contracts (DVN.sol, MultiSig.sol, DVNFeeLib.sol) are covered in the separate DVN Diff Audit Report.

EndpointV2 protocol contracts are covered in the separate EndpointV2 Diff Audit Report.

1.7 Summary

Contract Group	Changes (6ef8a4b -> 6356f0a)	Assessment
SendUln302, ReceiveUln302	License, pragma, error prefixes	No supplementary audit needed
UlnBase, SendUlnBase, ReceiveUlnBase	License, pragma, error prefixes	No supplementary audit needed
MessageLibBase, SendLibBase	License, pragma, error prefixes	No supplementary audit needed
SendLibBaseE2, ReceiveLibBaseE2	License, pragma, error prefixes	No supplementary audit needed
SendUln301, ReceiveUln301	License, pragma, error prefixes	No supplementary audit needed
SendLibBaseE1, ReceiveLibBaseE1	License, pragma, error prefixes	No supplementary audit needed
TreasuryFeeHandler	License, pragma, error prefixes	No supplementary audit needed
AddressSizeConfig, UlnOptions, DVNOptions	License, pragma, error prefixes	No supplementary audit needed
Worker	License, pragma, error prefixes	Covered in DVN report



2 Contract Verification Results

2.1 Diff Analysis: Final Audit (6ef8a4b) -> Target Commit (6356f0a)

Change Classification

All 18 in-scope contracts differ between 6ef8a4b and 6356f0a. Every change is cosmetic:

Contract File	Role in Audit	Verification Result
EndpointV2.sol	Main contract	Identical
MessageLibManager.sol	Dependency	Identical
MessagingChannel.sol	Dependency	Identical
MessagingComposer.sol	Dependency	Identical
MessagingContext.sol	Dependency	Identical

No function signatures, storage layouts, access control checks or business logic were modified.

Line Counts

Contract	Lines Changed
UlnBase	66
ReceiveUlnBase	63
ReceiveUln302	48
SendLibBase	38
ReceiveLibBaseE2	28
SendLibBaseE2	22
SendUln302	20
Worker	20
MessageLibBase	16
SendUlnBase	8

All diffs consist exclusively of the cosmetic changes listed in the previous table.



2.2 Diff Analysis: 6356f0a -> Current main

Identical

The following 10 contracts are byte-identical between 6356f0a and main:

MessageLibBase, SendUln301, ReceiveUln301, SendLibBaseE1, ReceiveLibBaseE1, TreasuryFeeHandler, AddressSizeConfig, UlnOptions, DVNOptions, Treasury

An additional 10 contracts (SendUln302, ReceiveUln302, UlnBase, SendUlnBase, ReceiveUlnBase, SendLibBase, SendLibBaseE2, ReceiveLibBaseE2, Worker, SafeCall) have also been confirmed identical between 6356f0a and main.

Changed (out of scope)

Three contracts differ on main but were **not** in the original MessageLib audit scope:

Contract	Changes on main
Executor.sol	Inlined ILayerZeroEndpointV2 interface, added lzRead support (nativeDrop, lzComposeAlert), added execute301 overload
ExecutorFeeLib.sol	Added lzRead fee support, ReadCmdCodecV1 integration
PriceFeed.sol	Added estimateFeeByEid function (alias for estimateFeeOnSend)

New files on main

13 new files exist on main that are absent at 6356f0a, including lzRead features (ReadLib1002, ReadLibBase, SimpleReadExecutor, LzExecutor), view helpers (ReceiveUln301View, ReceiveUln302View), lzRead codec/support types (ReadCmdCodecV1, SupportedCmdTypes) and ExecutorOptions. These are post-target additions and out of scope.



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